

Gastric Cancer Multi-Omics — Complete Results and Findings

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Scope. This document reports every quantitative result underlying the manuscript, together with its interpretation. Values are read programmatically from the analysis output files in `results/`; none are transcribed by hand. Negative and null results are reported alongside positive ones.

1. Prognostic signature — internal performance

LASSO-Cox signature (25 genes) trained on TCGA-STAD. Performance estimated by **nested** cross-validation, in which gene selection and penalty tuning occur only in the inner loop, so the outer-fold estimate is free of selection leakage.

metric	estimate	ci_low	ci_high	method
Harrell_C_ensemble	0.6112	0.562	0.6589	ensemble OOF LP, 2000x subject bootstrap
Uno_C_ensemble	0.5727	0.5246	0.6277	ensemble OOF LP, IPCW (timewt=n/G2), 2000x bootstrap
Harrell_C_perRepeat	0.5981	0.5709	0.6249	mean +/- 2.5-97.5 pctile over 20 repeats
Uno_C_perRepeat	0.5669	0.5411	0.5857	mean +/- pctile over 20 repeats
IntegratedBrierScore	0.1789	0.1732	0.1847	IPCW IBS to 5y, mean +/- pctile over repeats
Harrell_C_lambda1se_ensemble	0.5974			lambda.1se sensitivity, ensemble OOF LP
Harrell_C_lambda1se_perRepeat	0.5974	0.5974	0.5974	lambda.1se sensitivity, mean +/- pctile

Finding. Honest discrimination is modest: Harrell C = 0.611 (95% CI 0.562–0.659). The apparent (leaky) C on the same data was 0.72; the ~0.11 difference is selection optimism.

2. External validation (three independent cohorts, n = 922)

cohort	platform	n	events	genes_mapped	C_index	logrank_p	HR	HR_low	HR_high	HR_p
GSE84437	GPL6947	431	207	25	0.5297	0.4551	1.109	0.844	1.456	0.4585
ACRG/ GSE62254	GPL570	300	152	25	0.6079	8.632e-05	1.896	1.37	2.625	0.0001131
GSE15459	GPL570	191	95	25	0.5752	0.01365	1.676	1.107	2.537	0.01473

Finding. Validated in ACRG/GSE62254 and GSE15459; **failed** in GSE84437 (C = 0.530, HR 1.11, 95% CI 0.84–1.46, p = 0.46). GSE84437 is 89% pT3–T4, so it lacks the early-stage variation the signature detects.

2.1 Stage-stratified re-test in the failing cohort

subset	n	events	C	C_se	HR_perSD	CI	p	C_below_0.5
All (T1–T4)	431	207	0.465	0.021	1.14	1.00–1.29	0.0512	True
Early (T1–T3)	141	47	0.444	0.046	1.21	0.96–1.54	0.108	True
Late (T4)	290	160	0.483	0.024	1.09	0.93–1.27	0.29	True
T2–T3 only	130	45	0.442	0.047	1.23	0.96–1.56	0.0978	True

Finding. The non-validation persists in every stage stratum — reported as a clean negative.

2.2 Per-SD age/stage-adjusted estimates (meta-analysis inputs)

cohort	n	events	genes_mapped	adjustment	logHR	SE	HR	HR_low	HR_high	p
ACRG/ GSE62254	300	152	25	age + TNM stage	0.2633	0.08552	1.301	1.1	1.539	0.002083
GSE15459	191	95	25	age + TNM stage	0.182	0.1065	1.2	0.9736	1.478	0.08751
GSE84437	431	207	25	age + T-stage (no TNM stage in GEO; T-stage proxy)	0.104	0.0672	1.11	0.9726	1.266	0.1218

2.3 Random-effects meta-analysis (REML + Hartung-Knapp)

method	k	pooled_HR	CI_low	CI_high	p_value	I2	tau2	PI_low	PI_high	significant
REML + Hartung- Knapp (metafor rma, test=knha)	3	1.188	0.9616	1.466	0.07264	19.23	0.001718	0.9009	1.565	NO (CI crosses 1)

Finding. Pooled HR 1.19 (95% CI 0.96-1.47), $p = 0.073$, $I^2 = 19.2\%$. The confidence interval includes 1, so the pooled effect is **not statistically significant**.

3. Proportional-hazards assessment

model	term	chisq	df	p
signature	risk	9.778	1	0.001766
signature	GLOBAL	9.778	1	0.001766
signature+Stage+Age	risk	8.761	1	0.003077
signature+Stage+Age	Stage	2.982	3	0.3943
signature+Stage+Age	Age	1.843	1	0.1746
signature+Stage+Age	GLOBAL	13.93	5	0.01606

model	month	HR	CI_low	CI_high
tt log-time (age-adj, Stage-stratified)	12	1.488	1.227	1.804
tt log-time (age-adj, Stage-stratified)	36	1.029	0.8237	1.287
tt log-time (age-adj, Stage-stratified)	60	0.8674	0.6389	1.178

model	interval	HR	CI_low	CI_high	p
survSplit step (age-adj, Stage-stratified)	[0,12)m	1.986	1.45	2.72	1.9e-05
survSplit step (age-adj, Stage-stratified)	[12,36)m	1.099	0.8624	1.401	0.4448
survSplit step (age-adj, Stage-stratified)	[36,60)m	1.064	0.7072	1.602	0.7646
survSplit step (age-adj, Stage-stratified)	[60+)m	0.9436	0.4345	2.049	0.8834

Finding. The proportional-hazards assumption is violated (cox.zph $p = 0.0018$ signature alone; $p = 0.0031$ age/stage-adjusted). The signature is prognostic **early** (HR 1.49 at 12 months) and attenuates to null thereafter (1.03 at 36 months; 0.87 at 60 months). It is an early-risk marker, not a durable prognostic gradient.

4. Clinical added value over standard staging (external, ACRG)

statistic	value
deltaC_corrected(combined-clinical)	0.004538
LRT_chisq	9.508
LRT_df	1
LRT_p	0.002046
IDI@3y	0.021
IDI_p	0.04
continuousNRI@3y	0.175
NRI_p	0.04

Finding. $\Delta C = +0.0045$ over a clinical model; LRT p = 0.0020; IDI@3y 0.021; cNRI@3y 0.175. The gain is statistically detectable but **clinically negligible**; decision-curve analysis shows no incremental net benefit over staging.

5. Signature stability under resampling

metric	value
bootstraps_B	200
candidate_pool_size	1038
median_model_size	49
min_model_size	24
max_model_size	80
signature_genes_total	25
signature_in_candidate_pool	25
signature_selected_gt50pct	13
signature_selected_gt80pct	0
median_selection_freq_of_25	0.505
SERPINE1_freq	0.7
POSTN_freq	0.375
MATN3_freq	0.22

Finding. Across 200 bootstraps, 13 of 25 genes were selected in >50% of resamples and **none exceeded 80%**. The underlying programme is robust; the specific gene list is not.

6. Co-expression network analysis (WGCNA)

6.1 Module preservation in three external cohorts — primary finding

cohort	platform	moduleSize	red_genes_in_common	Zsummary.pres	medianRank.pres	strength
ACRG_GSE62254	GPL570 (ACRG curated)	77	77	15.85	2	STRONG (Z>10)
GSE15459	GPL570	233	233	16.8	3	STRONG (Z>10)
GSE84437	GPL6947	233	233	17.08	5	STRONG (Z>10)

6.2 Module eigengene survival association (external)

cohort	platform	n	events	red_genes_used	PC1_varExplained	HR_perSD	CI_low	CI_high	z	p
ACRG_GSE62254	GPL570 (ACRG curated)	300	152	77	0.506	1.274	1.091	1.487	3.068	0.002155
GSE15459	GPL570	191	95	233	0.331	1.548	1.248	1.919	3.983	6.813e-05
GSE84437	GPL6947	431	207	233	0.265	1.237	1.08	1.416	3.075	0.002107

Finding. The red (stromal/CAF) module is **strongly preserved** in all three cohorts (Zsummary 15.9, 16.8, 17.1; all >10) and independently prognostic in all three (HR/SD 1.24–1.55, all $p < 0.005$). Critically it replicates in GSE84437, where the 25-gene signature failed: the biology is more robust than the derived gene score.

7. Tumour-purity sensitivity

model	term	HR	lo	hi	p	n	ev
sig ~ score (all OS)	sig_score	2.606	2.136	3.179	3.689e-21	383	156
sig ~ score (purity subset)	sig_score	2.933	2.249	3.825	2.031e-15	250	92
sig ~ score + ABSOLUTE.purity	sig_score	2.965	2.276	3.862	7.585e-16	250	92
(purity coef in above)	purity	1.004	0.9956	1.013	0.3507	250	92
sig ~ score + xCell.Stroma	sig_score	2.757	2.235	3.402	3.092e-21	383	156
sig ~ score + purity+stage+age	sig_score	3.58	2.584	4.961	1.791e-14	209	68
(purity in full model)	purity	1.009	0.9992	1.02	0.07189	209	68
caf ~ score (all OS)	caf_score	1.305	1.112	1.531	0.001093	383	156
caf ~ score (purity subset)	caf_score	1.333	1.058	1.678	0.0146	250	92
caf ~ score + ABSOLUTE.purity	caf_score	1.342	1.052	1.712	0.01783	250	92
caf ~ score + xCell.Stroma	caf_score	1.267	1.042	1.542	0.01781	383	156
caf ~ score + purity+stage+age	caf_score	1.532	1.128	2.082	0.006337	209	68
ref: ABSOLUTE.purity alone	purity	0.9969	0.9881	1.006	0.4861	250	92
ref: xCell.Stroma alone	stroma_x	6.015	1.439	25.15	0.01396	383	156

Finding. The signature survived adjustment for ABSOLUTE-estimated tumour purity (HR 2.97, 95% CI 2.28–3.86, $p = 7.6 \times 10^{-16}$), with purity itself non-significant ($p = 0.35$). The prognostic signal is stromal biology, not a sampling artefact.

8. Immune deconvolution validated against measured pathology

estimate	measured	n	spearman_rho	p
CD8 T cells (MCP)	Leukocyte %	272	0.4684	3.067e-16
CD8 T cells (MCP)	Lymphocyte infiltration %	274	0.01981	0.7441
T cells (MCP)	Leukocyte %	272	0.6656	3.566e-36
T cells (MCP)	Lymphocyte infiltration %	274	-0.007241	0.905
CD8+ T-cells (xCell)	Leukocyte %	272	0.3804	8.511e-11
CD8+ T-cells (xCell)	Lymphocyte infiltration %	274	0.03785	0.5328
ImmuneScore (xCell)	Leukocyte %	272	0.6525	2.23e-34
ImmuneScore (xCell)	Lymphocyte infiltration %	274	-0.05126	0.398

Finding. Predicted T-cell content correlates with pathologist-measured leukocyte percentage (Spearman $\rho = 0.666$, $p = 3.6 \times 10^{-36}$, $n = 272$). Validation is measure-specific: CD8 estimates track leukocyte percentage ($\rho =$

0.468) but **not** lymphocyte-infiltration percentage ($p = 0.020$, $p = 0.74$), so deconvolution captures overall immune burden rather than lymphocyte subset composition.

9. Single-cell localisation of the prognostic module

cell_type	n_cells	pct
Epithelial	30195	68.64
Endothelial	3357	7.63
Myeloid	2809	6.39
Plasma	2571	5.84
Tcell	2534	5.76
Fibroblast	1828	4.16
Mast	425	0.97
Bcell	273	0.62

9.1 Non-circular hub-gene localisation

Genes used to annotate clusters were **excluded** before testing, to avoid circular reasoning.

gene	dominant_cell_type	frac_in_dominant	panel
ANTXR1	Fibroblast	0.863	stromal_hub
COL6A3	Fibroblast	0.977	stromal_hub
SPARC	Fibroblast	0.653	stromal_hub
POSTN	Fibroblast	0.973	stromal_hub
MXRA8	Fibroblast	0.98	stromal_hub
FN1	Fibroblast	0.914	stromal_hub
COL8A1	Fibroblast	0.998	stromal_hub
VCAN	Fibroblast	0.972	stromal_hub
GFPT2	Fibroblast	0.527	stromal_hub
THBS2	Fibroblast	0.82	stromal_hub
AEBP1	Fibroblast	0.924	stromal_hub
BGN	Fibroblast	0.972	stromal_hub
SULF1	Fibroblast	0.773	stromal_hub
COL5A1	Fibroblast	0.937	stromal_hub
BICC1	Fibroblast	0.97	stromal_hub
FBN1	Fibroblast	0.932	stromal_hub
ISLR	Fibroblast	0.902	stromal_hub
CDH11	Fibroblast	0.96	stromal_hub
SFRP4	Fibroblast	0.987	stromal_hub
SCARF2	Fibroblast	0.736	stromal_hub
PRRX1	Fibroblast	0.979	stromal_hub
FNDC1	Fibroblast	0.983	stromal_hub
ITGA11	Fibroblast	0.984	stromal_hub

Finding. All **23 of 23** remaining hub genes are fibroblast-dominant (median fraction 0.960). Independent, non-circular confirmation that the prognostic module is fibroblast-derived.

10. Tumour microbiome

10.1 Reprocessing fidelity

stage	pearson_r	n_samples_matched
pooled_all_stages	0.9833	932
input	1	932
filtered	0.9678	932
denoised	0.9683	932
merged	0.9692	932
nonchim	0.9682	932

DADA2 reprocessing of 944 raw libraries reproduced the original study's published read-tracking table at Pearson $r = 0.983$.

10.2 Alpha diversity across the Correa cascade

metric	median_Nonul	median_GCN	cliffs_delta	wilcox_p
Observed	47	30	-0.27	3.3e-07
Shannon	2.176	1.923	-0.122	0.0211
Simpson	0.781	0.756	-0.072	0.172

Finding. Observed richness falls (median 47 $\rightarrow$ 30; Cliff's $\delta = -0.27$, $p = 3.3 \times 10^{-7}$) and Shannon declines modestly ($p = 0.021$), but **Simpson does not change significantly** ($p = 0.172$). Rare taxa are lost while dominant community structure is preserved.

10.3 Batch-confounding audit — critical negative

Unnamed: 0	KVF37	L2HBK	L3RVN	L3RVR	L7Y62	L848P	LJDKG
Non-ul	0	11	0	72	67	0	144
Ul	0	0	0	1	28	0	74
GCN	0	0	0	22	78	1	103
GCT	1	0	95	0	0	200	0

metric	value
cancer_vs_control_AUC	0.9161
cancer_AUC_CI_lo	0.896
cancer_AUC_CI_hi	0.9362
flowcell_pred_accuracy	0.7759
flowcell_majority_baseline	0.545
flowcell_multiclass_AUC	0.7306

Finding. A classifier separates cancer from control at AUC 0.916 — but the same data predicts **sequencing flowcell** at 77.6% accuracy against a 54.5% majority baseline, and flowcell adjustment collapses the community difference (Bray-Curtis R^2 0.065 $\rightarrow$ 0.011). Tumour and control samples occupy near-disjoint flowcells, so batch and biology are statistically inseparable. The apparent dysbiosis is substantially a **batch artefact**.

10.4 Differential abundance (paired: tumour vs same patient's adjacent normal)

genus	effect_clr_paired	p	q
Serinicoccus	-2.732	8.591e-11	5.24e-09
Bacteroides	-0.7256	7.198e-10	2.195e-08
Dietzia	-1.93	1.102e-08	2.24e-07
Faecalibacterium	-0.4796	1.872e-06	2.351e-05
Pseudomonas	-0.4094	1.927e-06	2.351e-05
Helicobacter	-1.455	3.172e-06	3.225e-05
Microbacterium	-0.298	9.879e-05	0.0008609
Blautia	-0.3692	0.0001448	0.001104
Gemella	0.4311	0.0002222	0.001506
Sphingomonas	0.303	0.000447	0.002727
Fusobacterium	0.5201	0.0005137	0.002849
Streptococcus	0.4914	0.003322	0.01689
Leptotrichia	0.2251	0.005212	0.02354
Prevotella_7	0.2924	0.005404	0.02354
Brachybacterium	-0.3197	0.008071	0.03282

Finding. Of 61 genera tested, 18 were significant at $q < 0.05$. The largest effects are a **mixture** of two kinds of taxon. Two are environmental/actinobacterial organisms consistent with the batch artefact identified in §10.3 (Serinicoccus -2.73 , $q = 5.2 \times 10^{-9}$; Dietzia -1.93 , $q = 2.2 \times 10^{-7}$). The remainder, however, are genuine gastric, gut and oral flora: **Helicobacter** is strongly depleted in tumour tissue (-1.45 , $q = 3.2 \times 10^{-5}$), alongside Bacteroides (-0.73), Faecalibacterium (-0.48), Pseudomonas (-0.41) and Blautia (-0.37), while Fusobacterium ($+0.52$), Streptococcus ($+0.49$) and Gemella ($+0.43$) are enriched.

Interpretation. The Helicobacter depletion with oral-taxon enrichment is the pattern predicted by the Correa cascade: as atrophic gastritis destroys acid-producing parietal cells, the stomach becomes hospitable to oral and gut organisms that could not previously survive there. This paired analysis therefore contains real biology **as well as** batch signal — which is precisely why the flowcell audit in §10.3 is necessary: the two cannot be separated in this cohort, so neither can be claimed cleanly. Note that the contaminant genera driving the classifier in §10.3 are not the same set as the top paired-DA hits (Serratia $q = 0.69$ and Cutibacterium $q = 0.92$ are both non-significant here).

11. Mendelian randomisation — causal assessment

11.1 Instrument strength

Exposure	p_threshold	nSNP_prclump	nSNP_used	mean_F	min_F	IVW_OR	IVW_CI	IVW_p	PRESSO_global_p
H. pylori IgG seropositivity	5e-8 -> 1e-5 (locus-wide)	21	17	21.6	19.7	0.96	0.71-1.30	0.79	0.052
Streptococcus (genus)	5e-8 -> 1e-5 (locus-wide)	17	15	22.5	19.3	1.1	0.90-1.34	0.35	0.159
Fusobacterium	5e-8 -> 1e-5 (locus-wide)	25	23	22.8	19.6	1.04	0.79-1.36	0.79	0.611
Prevotella	5e-8 -> 1e-5 (locus-wide)	19	15	20.9	19.4	0.98	0.84-1.14	0.76	0.605
Veillonella	5e-8 -> 1e-5 (locus-wide)	11	8	21.2	19.9	1.04	0.84-1.29	0.69	0.676
Lactobacillus	5e-8 -> 1e-5 (locus-wide)	12	10	22.1	20.3	0.96	0.84-1.09	0.51	0.542

11.2 Primary IVW estimates

exposure	nsnp	b	se	pval
H. pylori IgG seropositivity	17	-0.04234	0.1556	0.7855
Streptococcus (genus)	15	0.09506	0.1014	0.3483
Fusobacterium	23	0.03788	0.1389	0.7851
Prevotella	15	-0.02362	0.07849	0.7634
Veillonella	8	0.04294	0.1083	0.6917
Lactobacillus	10	-0.04231	0.06446	0.5115

11.3 Sensitivity: MR-Egger pleiotropy

id.exposure	id.outcome	outcome	exposure	egger_intercept	se	pval
ebi-a-GCST90006910	ebi-a-GCST90018849	Gastric cancer id:ebi-a-GCST90018849	H. pylori IgG seropositivity	-0.008465	0.01966	0.673
ebi-a-GCST90017070	ebi-a-GCST90018849	Gastric cancer id:ebi-a-GCST90018849	Streptococcus (genus)	-0.002894	0.03305	0.9315
ebi-a-GCST90032406	ebi-a-GCST90018849	Gastric cancer id:ebi-a-GCST90018849	Fusobacterium	0.03821	0.01747	0.04018
ebi-a-GCST90017045	ebi-a-GCST90018849	Gastric cancer id:ebi-a-GCST90018849	Prevotella	-0.009258	0.03218	0.7781
ebi-a-GCST90017088	ebi-a-GCST90018849	Gastric cancer id:ebi-a-GCST90018849	Veillonella	0.1468	0.07943	0.114
ebi-a-GCST90017030	ebi-a-GCST90018849	Gastric cancer id:ebi-a-GCST90018849	Lactobacillus	-0.03954	0.02119	0.09902

11.4 Sensitivity: MR-PRESSO global test

exposure	RSSobs	PRESSO_global_p
H. pylori IgG seropositivity	32.6	0.052
Streptococcus (genus)	23.07	0.159
Lactobacillus	10.29	0.542
Prevotella	13.97	0.605
Fusobacterium	21.78	0.611
Veillonella	6.394	0.676

Finding. All six microbial exposures gave **null** IVW estimates against gastric cancer (smallest $p = 0.348$, Streptococcus). Instruments were adequately strong ($F = 19.3$ – 22.8). MR-PRESSO global tests were non-significant for all six ($p = 0.052$ – 0.676). One MR-Egger intercept was nominally significant (Fusobacterium 0.038 , $p = 0.040$) and is disclosed. **Interpretation:** at current instrument and outcome power (1,029 cases / 475,087 controls) these data provide **no evidence of a causal microbial effect** — which is not the same as evidence of no effect.

12. Therapeutic target dependency (DepMap CRISPR, 35 gastric lines)

gene	target_class	n_gastric	gastric_mean_CERES	gastric_median	gastric_min	gastric_frac_dep	pancancer_mean	pancancer_min
PIK3CA	PI3K/mTOR	35	-0.742	-0.5974	-2.141	0.5714	-0.4859	0.39
MTOR	PI3K/mTOR	35	-1.184	-1.123	-2.284	1	-1.172	0.98
PIK3CB	PI3K/mTOR	35	-0.1891	-0.151	-0.8599	0.05714	-0.195	0.06
CDK4	CDK4/6	35	-0.8248	-0.5964	-3.023	0.6571	-0.7646	0.66
CDK6	CDK4/6	35	-0.5485	-0.5352	-1.265	0.5143	-0.5882	0.50
FGFR1	FGFR	35	-0.0375	-0.0009924	-0.7769	0.02857	-0.1232	0.09
FGFR2	FGFR	35	-0.2204	-0.03647	-2.588	0.1143	-0.03131	0.01
FGFR3	FGFR	35	0.06172	0.06064	-0.2086	0	0.06429	0.00
FGFR4	FGFR	35	0.05795	0.06845	-0.5669	0.02857	0.03188	0.00

Finding. PIK3CA is both dependent and **selective** for gastric lines (gene effect -0.742); MTOR (-1.184) is common-essential; CDK4 (-0.825) and CDK6 (-0.548) are dependencies. **FGFR1-4 are not dependencies** (-0.220 to $+0.062$; FGFR3 and FGFR4 score positive, i.e. no dependency at all), despite FGFR inhibitors appearing among repurposing hits.

13. Benchmark: previously published 5-gene gastric signature

n	events	C_apparent	C_optimism_corrected	KM_logrank_p	HR_highVsLow
383	156	0.5446	0.4807	0.0039	0.627

Finding. Re-run on identical data with identical optimism correction, a prior published 5-gene signature achieved $C = 0.481$ — **below chance** (apparent C 0.545). This is the context in which the present signature's honest C of 0.611 should be read.

14. Pathway enrichment

pathway	NES	padj
HALLMARK_E2F_TARGETS	3.751	9.555e-54
HALLMARK_G2M_CHECKPOINT	3.642	2.221e-47
HALLMARK_MYC_TARGETS_V1	2.669	2.013e-16
HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION	1.85	1.448e-05
HALLMARK_OXIDATIVE_PHOSPHORYLATION	-2.542	1.542e-14
HALLMARK_FATTY_ACID_METABOLISM	-2.579	2.038e-13

14.1 Diffuse versus intestinal (Lauren) contrast

pathway	NES	padj
HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION	3.174	1.58e-40

Finding. Tumours show coordinated up-regulation of proliferation programmes (E2F targets NES $+3.75$; G2M checkpoint $+3.64$; MYC targets $+2.67$) with down-regulation of oxidative phosphorylation (-2.54) and fatty-acid metabolism (-2.58) — the transcriptional signature of proliferative reprogramming. EMT is the top Hallmark set in diffuse tumours (NES $+3.17$, padj 1.6×10^{-40}), independently recovering the classical Lauren histological distinction from expression alone.

15. Summary of findings

Finding	Status	Key evidence
Stromal/CAF programme underlies prognosis	Robust, replicated	Module preserved in 3/3 cohorts (Z 15.9–17.1); prognostic in 3/3 (HR/SD 1.24–1.55); 23/23 hub genes fibroblast-dominant
25-gene prognostic signature	Modest, bounded	Nested-CV C 0.611; validates 2/3 cohorts; pooled HR 1.19 (0.96–1.47) not significant; early-acting only
Clinical utility over staging	Negligible	ΔC +0.0045; no incremental net benefit on decision-curve analysis
Proliferation up / oxidative metabolism down	Confirmed	E2F +3.75, G2M +3.64, MYC +2.67; OXPHOS –2.54, FAO –2.58
EMT specific to diffuse subtype	Confirmed	NES +3.17, padj 1.6×10^{-40}
Immune deconvolution validity	Partly validated	ρ 0.666 vs measured leukocyte %; but ρ 0.020 vs lymphocyte-infiltration %
Tumour-microbiome dysbiosis	Batch artefact	78% flowcell-predictable vs 55% baseline; Bray R ² 0.065→0.011 on adjustment; contaminant-led classifier
Reduced microbial diversity	Partly replicated	Richness 6 –0.27 (p 3.3e-7); replicates in Portugal, null in Italy; Simpson ns
Causal microbial effect on gastric cancer	No evidence	All six MR exposures null; smallest p 0.348; underpowered (1,029 cases)
PI3K/mTOR and CDK4/6 dependency	Experimentally supported	PIK3CA –0.742 (selective), MTOR –1.184, CDK4 –0.825, CDK6 –0.548
FGFR dependency	Not supported	FGFR1–4 gene effect –0.220 to +0.062, none dependent
Prior published 5-gene signature	Fails on same data	Optimism-corrected C 0.481 (below chance)